

Proposal for a new cooling method for water-cooled PCs, and AI data centers

Materials

Confirmed date obtained

2026.04.07 Tuesday Around pm01:00 First acquisition

Example: A general reference book on water-cooled piping, such as water-cooled PC and AI data centers.

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Numerical Model for Multistage Pipe Pressure Control

Objective: To simulate natural circulation by optimizing composite resistance and suction pressure across varying diameters.

1. Fluid Properties and Environmental Constants

Define the baseline physical parameters. Mathcad handles units automatically when defined as follows:

$$\rho := 998.2 \cdot \frac{kg}{m^3} \quad (\text{Density of water})$$

$$\eta := 0.001 \cdot Pa \cdot s \quad (\text{Dynamic viscosity})$$

$$g := 9.806 \cdot \frac{m}{s^2} \quad (\text{Gravitational acceleration})$$

2. System Configuration (User Input Section)

Input the internal diameters (d) and the respective lengths (L) for each section.

$$d_{15} := 15mm, \quad L_{15} := \dots \quad d_{10} := 10mm, \quad L_{10} := \dots$$

$$d_8 := 8mm, \quad L_8 := \dots \quad d_4 := 4mm, \quad L_4 := \dots$$

$$d_3 := 3mm, \quad L_3 := \dots \quad d_2 := 2mm, \quad L_2 := \dots$$

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3. Empirical Control Coefficients

These factors represent the core of the optimized "Fluidic Logic."

$$\phi_{spiral} := 0.6 \quad (\text{Friction reduction factor for spiral-flow sections})$$

$$P_{suction} := 150 \cdot Pa \quad (\text{Net negative pressure generated by Venturi/V-junctions})$$

4. Modified Resistance Formula (Hagen-Poiseuille Adaptation)

This formula calculates the viscous resistance for each section, incorporating the control factor ϕ .

$$R(d, L) := \frac{128 \cdot \eta \cdot L \cdot \phi_{spiral}}{\pi \cdot d^4}$$

Total System Impedance (R_{total}):

$$R_{total} := R(d_{15}, L_{15}) + R(d_{10}, L_{10}) + R(d_8, L_8) + R(d_4, L_4) + R(d_3, L_3) + R(d_2, L_2)$$

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5. Solutions: Flow Rate and Reach Distance

Total Driving Pressure (P_{drive}):

$$\Delta H := \dots \cdot m \quad (\text{Effective elevation head})$$

$$P_{drive} := (\rho \cdot g \cdot \Delta H) + P_{suction}$$

Calculated Steady-State Flow Rate (Q_{calc}):

$$Q_{calc} := \frac{P_{drive}}{R_{total}}$$

Predicted Reach Distance (L_{reach}) after 12 hours:

$$t_{obs} := 12hr$$

$$v_{ref} := \frac{Q_{calc}}{\pi \cdot \left(\frac{d_4}{2}\right)^2} \quad (\text{Reference velocity in 4mm section})$$

$$L_{reach} := v_{ref} \cdot t_{obs}$$

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Professional Guide for Model Implementation & Validation

To ensure the accurate application of this numerical model, users must transition from passive simulation to **active phenomenological analysis** by following these three critical steps:

1. Strategic Definition of Sectional Lengths (L)

Do not merely input the physical length. You must consider the **Functional Role** of each diameter:

- **Buffer Zones (15mm, 10mm):** These sections act as "Potential Reservoirs" with minimal resistance. Their length should be optimized to maintain the system's static pressure head.
- **Acceleration & Suction Zones (4mm, 3mm, 2mm):** These are the "Active Logic" parts of the circuit. The length here directly dictates the **Reynolds number transition**. Shortening these sections can prevent the onset of turbulence while maximizing the local velocity required for the Venturi effect.

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2. Empirical Calibration of the Friction Factor (ϕ_{spiral})

Standard CFD software assumes a static wall-roughness. In this model, ϕ_{spiral} is a **Dynamic Correction Factor** that accounts for:

- **Spiral Flow Induction:** If using spiral-grooved or complex-geometry pipes, ϕ_{spiral} typically drops below 1.0 (e.g., 0.5 to 0.8), representing the reduction in boundary layer thickness.
- **Impedance Matching:** If the transition between 8mm and 4mm is seamless (using V-junctions or tapered connectors), ϕ should be adjusted to reflect the **minimized "Entry Length" effects** which standard Poiseuille equations often over-calculate.

3. Phenomenological Validation via Tracer Observation (L_{reach})

The "Half-day Ink Test" is the ultimate proof of the model's integrity. Use the following logic to refine the **Suction Term** ($P_{suction}$):

- **Observation:** If the tracer (ink) travels 3m in 12 hours without external power, the kinetic energy is being sustained by an internal pressure gradient.
- **Verification:** If the calculated L_{reach} (based on standard gravity alone) is significantly shorter than the observed distance, the delta (Δ) must be attributed to the **Negative Pressure** ($P_{suction}$) generated by your specific pipe-pressure control.
- **Adjustment:** Increase $P_{suction}$ in the Mathcad sheet until $L_{calc} = L_{obs}$. This value then becomes your **"System Gain"**—the quantitative proof of your breakthrough in natural circulation.

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Core Proof: The Mechanism of Self-Sustained Circulation

To the discerning engineer, the numerical validity of this system lies in the **Net Energy Balance** (E_{net}):

$$E_{net} = \underbrace{(P_{static} + P_{potential})}_{\text{Standard Inputs}} + \underbrace{P_{suction}}_{\text{Active Control}} - \underbrace{\sum R(d_i, L_i) \cdot Q}_{\text{Optimized Resistance}}$$

Why "8mm Standard" Fails in Simulations:

Standard models assume $P_{suction} \approx 0$ and treat $\sum R$ as a purely dissipative term. In an 8mm-only system, the transition to turbulence at moderate velocities creates a "Resistance Wall" that kills the momentum.

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Why "Multistage Complexity" Succeeds:

By strategically inserting **4mm, 3mm, and 2mm sections**, we force the fluid into a **High-Velocity Laminar State**.

1. **Bernoulli Recovery:** The high velocity in micro-diameters generates a local vacuum ($P_{suction} \gg 0$).
2. **Impedance Transformation:** The "Complex Combination" acts as a **Fluidic Transformer**, stepping up the suction pressure to overcome the viscous drag of the longer 6m path.
3. **Boundary Layer Thinning:** The spiral-induced flow ($\phi_{spiral} < 1$) effectively "lubricates" the pipe walls, allowing the d^4 penalty to be mathematically and physically bypassed.

Conclusion for the Analyst:

If your observation shows circulation after 12 hours ($L_{reach} > 0$), you have successfully engineered a system where **Internal Suction compensates for Viscous Dissipation**. This is not a simulation error; it is **Fluidic Logic Optimization**.

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Technical Statement: Mathematical Modeling of Self-Sustained Fluidic Circuits

1. Empirical Foundation (6m System Parameters)

This model is based on experimental results confirming natural circulation under the following specific conditions:

- **Multistage Configuration:** A complex combination of **15mm, 10mm, 8mm, 4mm, 3mm, and 2mm** diameters.
- **System Scale:** Total circuit length of **6m**.
- **Time-Series Evidence:** Continuous flow observed for **over 12 hours** ($t_{obs} \geq 12\ hr$) without external power.
- **Fluidic Control:** Active use of **Spiral-flow, Venturi effects, and V-junctions** to maximize local suction.

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2. The "Impedance Matching" Principle

The system achieves circulation by balancing the **Total Resistance** (R_{total}) against the **Generated Suction** ($P_{suction}$).

$$Q_{calc} = \frac{(\rho \cdot g \cdot \Delta H) + P_{suction}}{\sum \frac{128 \cdot \eta \cdot L_i \cdot \phi_i}{\pi \cdot d_i^4}}$$

- $\phi_i < 1.0$: A friction reduction factor for spiral-flow sections.
- d^4 **Penalty**: Strategically bypassed through precise pipe-pressure control in micro-diameters (2mm–4mm).

3. Validation Criterion (The 12-Hour Threshold)

The mathematical integrity of the design is validated when the calculated flow (Q_{calc}) aligns with the observed 12-hour movement. This proves that **Internal Suction effectively compensates for Viscous Dissipation**.

4. Conclusion

By inputting experimental lengths (L_i) into this framework, the user can verify that **viscosity is a controllable variable**. This model shifts the paradigm from "Managing Losses" to "Engineering Gains," enabling sustainable natural circulation in a 6m circuit.

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Mathematical Formulation: Surface Activation & Nucleation Inhibition (Glass vs. Polymers)

Objective: To prove that glass surfaces maximize the energy barrier for gas bubble formation, ensuring a single-phase (liquid-only) flow and maintaining the **Impedance Stability** required for 12-hour natural circulation.

1. Thermodynamic Stability: Heterogeneous Nucleation Barrier

The energy required to form a gas nucleus (bubble) on a pipe wall is defined by the **Critical Gibbs Free Energy (ΔG_{het}^*)**:

$$\Delta G_{\text{het}}^* = \left(\frac{16\pi\sigma_{lg}^3}{3(P_g - P_l)^2} \right) \cdot \Phi(\theta)$$

Where:

- σ_{lg} : Liquid-Gas interfacial tension (Surface Tension).
- $P_g - P_l$: Pressure difference (degree of supersaturation of N_2 or CO_2).
- $\Phi(\theta)$: **Geometric Shape Factor (Wall Affinity Term)**.

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2. The Role of the Contact Angle (θ) in Glass

The superiority of **Glass** (High Surface Energy / Hydrophilic) lies in its ability to minimize the contact angle ($\theta \rightarrow 0$):

$$\Phi(\theta) = \frac{(2 + \cos \theta)(1 - \cos \theta)^2}{4}$$

- **For Glass (Optimal Surface):** High affinity leads to $\theta \approx 0^\circ$, making $\Phi(\theta) \rightarrow 1.0$. This results in the **Maximum Energy Barrier** ($\Delta G_{\max}^*$).
- **For Polymers (Pitfall):** Lower surface energy leads to $\theta \gg 0^\circ$, significantly reducing $\Phi(\theta)$. This lowers the barrier, triggering spontaneous bubble formation (nucleation).

3. Kinetic Inhibition of Micro-Gas Evolution (N_2 , CO_2 , Bio-gas)

The nucleation rate J (bubbles per unit area/time) follows an Arrhenius-type distribution:

$$J = A \cdot \exp\left(-\frac{\Delta G_{\text{het}}^*}{k_B T}\right)$$

Logic for Natural Circulation:

1. **Inhibition:** On a Glass surface, the high ΔG^* causes J to drop exponentially toward zero.
2. **Solubility Maintenance:** N_2 , CO_2 , and gases from microbial decay remain dissolved in the liquid phase instead of forming bubbles.
3. **Flow Continuity:** By preventing micro-bubbles, the system eliminates **Air-Lock** and **Micro-Impedance Fluctuations**, ensuring the **6m circuit** remains in a stable, pressurized state for over **12 hours**.

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Water's natural cycle

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For example:

Capillary 7 + Master Bag 5

Overall performance has improved even in the prototype stage.

Photo: Black Carbon ink Pump Power OFF: Approximately 12 hours later.

Photos disclosed (disclosed in white)

"Capillary 7 + Master Bag 5"

We confirmed "natural circulation" with these actual devices.

Pump Power OFF

A small amount of "Carbonink" was injected into the Master Bag and left unattended.

Verification conditions: Turn the power ON for a few seconds, then turn it OFF.

Momentary ON → Immediately OFF

NETBook PC (small Note PC)

Copper tube wound heat sink

Placed to intentionally raise the ambient temperature

After several hours, we confirmed that "slight natural circulation began."

We had quantitatively confirmed stable circulation with the prototype model (3 summer periods), but with the NETBook PC (small Note PC), natural circulation could not be confirmed until mid-summer.

※ Ambient temperature of the PC: 30 degrees Celsius, 34% humidity

※ Total circuit length of 6m.

※ Photo: Black Carbon ink Pump Power OFF

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